

Comparative Analysis of Classification Models for Heart Disease Prediction using Machine Learning

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Abstract— Among the most common health problems of modernity is heart disease, where early diagnosis becomes crucial for effective treatment and prevention. One cardinal responsibility of health professionals is the identification of at-risk persons to help decrease mortality through timely interventions. In this regard, a comparative analysis of machine learning classification models for heart disease prediction using a dataset with 13 medically relevant features such as gender, blood pressure, cholesterol level, family history, smoking habits, diet, treatment history, obesity, hypertension, and physical inactivity can be found in this study. The main objective of this research work is to find the most accurate and efficient classification model to predict the heart disease infallibly. Machine learning is a significant contribution to healthcare because it provides vast amounts of medical data that may be practically analyzed to derive actionable insights for clinical decision-making. Support Vector Machine (SVM) and Random Forest are the special focus among all examined models. It is found from the results that both models are doing very well in terms of predicting heart disease but Random Forest proved itself as a superior model as it achieved the highest accuracy in this study.

Keywords— Heart Disease Prediction, Machine Learning, Classification Models, Comparative Analysis, Random Forest, Support Vector Machine (SVM), Early Diagnosis, Medical Data, Healthcare Analytics, Clinical Decision-Making, Risk Identification, Predictive Modeling, Health Informatics, Accuracy Evaluation, Preventive Healthcare

I. INTRODUCTION

In consideration of being one of the foremost health concerns present in the world, heart diseases account for the relatively high percentage of 31 above all types of diseases and death. Among the contributing factors are lack of exercise or activity, poor nutrition, stress, and inborn factors. Decreasing mortality and improving patient management hinges on early detection and accurate diagnosis. However, common approaches to diagnosis that include electrocardiograms (ECGs) and stress tests often require large and costly

equipment coupled with a specialized physician's focused time, limiting access to timely interventions. The health field now seems to utilize machine learning (ML) more as a powerful tool for the analysis of large medical datasets to disclose hidden patterns so as to enhance diagnostic accuracy. Among the various machine learning techniques, classification models like Random Forest (RF) and Support Vector Machine (SVM) provide strong predictive capabilities for heart disease detection. Random Forest has been appreciated for the ability to handle complex datasets based on assessment of feature importance, thereby decreasing overfitting. SVM on the other hand is appreciated for its ability to discriminate between at-risk and non-risk patients in the highly dimensional data spaces. The majority of the existing studies on heart disease prediction by means of machine learning are affected by limitations involving small datasets, limited features, and unavailability of comparative model analysis. Besides, most of them concentrate only on the accuracy of their model and put very little knowledge on its computational efficiency and applicability in real-world settings. A handful even consider model scalability and integration with real-time health data. This study bridges those gaps by studying both predictive efficiency and computation cost between Random Forest and SVM using a medically relevant feature set. This research puts forward the comparative performance analysis of classification models for heart disease prediction. It evaluates RF and SVM models with the purpose of pinpointing the one that is able to make the most accurate diagnosis. The results obtained in this work lead to further data-driven healthcare solutions that would aid medical experts in making decisions and, thus, in the area of early detection and preventive care.

II. LITERATURE SURVEY

Coronary heart disease (CHD) is one of the greatest critical factors causing illness and death around the world, demanding great efforts at rapid diagnosis and risk evaluation

in order to improve the outcome of patients. In this study, the various machine learning (ML) algorithms for CHD prediction are discussed. The database under study comprises factors such as age, cholesterol, blood pressure, smoking habits, and family history of heart diseases most affecting the health status. Logistic regression, decision trees, random forest, support vector machines (SVM), and neural network techniques are applied to assess the efficacy of predicting the potential risk of heart disease.[2] heart disease prediction methodologies in machine learning have been reviewed, wherewith contrasting techniques have been reviewed to predict the heart's disease risk differing health parameters. These parameters include age, sex, cholesterol, blood pressure, smoking habits, exercise, and family history. Among other prevalent machine-learning models, Logistic regression models may be generally applied for calculating the probability of heart disease and thereby helps in categorizing the patients as being at risk or not depending on their health and lifestyle features.[3] It's a prediction analysis framework on heart disease by machine learning techniques to find the most important parameters in predicting whether a patient would develop the disease. The framework uses application of different machine learning models like logistic regression, random forests, SVMs, and neural networks to generate insights with intra-critical health parameters such as age, cholesterol, blood pressures, and family history.[4] coronary heart disease (CHD) is sometimes one of those health issues that need much worse thick-to-skin argument and too little good reaction to it. Immediate detection and proper treatment can make a big difference in survival. This work proposes the use of machine learning algorithms for early detection of CHD using Cleveland dataset patient data on cardiovascular risk factors such as age, sex, blood pressure, cholesterol levels as well as ECG result and exercise-induced chest pain. For this study, several classification techniques were used such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine, K-nearest Neighbors. Furthermore, various data preprocessing techniques such as data cleaning, normalization, and feature selection were applied to enhance the performance of the models being used. Besides that, it was found through literature review that preprocessing plays a significant role in improving accuracy prediction models of heart disease.[5] heart diseases are ever a big problem affecting human beings, a situation that warrants early diagnosis and prognosis so that health outcomes may be improved. The machine-learning techniques are being applied to predict heart disease on the basis of various clinical parameters such as person's age, cholesterol levels, blood pressure readings, electrocardiographic findings, and family history.[6] Detection of heart disease at the early point of the disease is vital in the healthcare field as it increases the chances of effective treatment and decreases mortality rates. The present study looks at the feasibility of using machine

learning models for predicting heart disease using biomedical data, which would include clinical parameters such as age, sex, blood pressure, cholesterol levels, electrocardiographic results, and exercise-induced angina.[7] The research's focus is on developing a hybrid machine learning system that enhances heart disease prediction with respect to both accuracy and reliability. The hybridization includes various machine learning classification algorithms and related ensemble techniques for enhanced prediction performance namely support vector machines (SVM), random forest, K-nearest neighbors, XGBoost, etc. and stacking models.[8] heart diseases, one of the major causes of global mortality, require timely and accurate diagnosis to design proper treatment and preventive measures. Preprocessing techniques such as handling missing values, feature scaling, data cleaning, and more were conducted to derive maximum benefits using machine learning. Subsequently, the performance evaluation of each model happened through standard evaluation metrics of accuracy, precision, recall, F1-score, and Area Under the Curve (AUC) calculations. Most models yielded fairly accurate results; however, Random Forest and Support Vector Machine (SVM) scored the highest accuracy levels, hence they are very useful and effective at detecting heart diseases from digital healthcare systems. [9] The pandemic brought about both destruction and rebirth in heart disease mechanisms. Previous prediction mechanisms gave a lot of importance to early risk stratification and improved assortment of patient care. The present survey offers a special, number-intensive study on the prediction of heart diseases using different machine-learning models, the advantages versus disadvantages, and their applications. The present study tests different algorithms vis-à-vis their predictability regarding heart disease risk factors, including Logistic Regression, Decision Trees, Random Forest, SVM, etc., KNN, ANNs, and ensemble approaches.[10] A new study has consequently provided a machine learning-based technique of early-heart disease prediction using Cleveland and Stat log datasets. Out of the six tested ML algorithms, SVM proved to be the best resulting in accuracy of up to 92% after feature selection and hyperparameter tuning with 10-fold cross-validation. This model was further validated on Stat log dataset giving an accuracy of 91.18% thus strengthening the result. The research demonstrates the worthiness of using ML in early assessment of cardiovascular risks, emerging as a useful tool for health practitioners.

III. EXISTING SYSTEM

Cardiovascular diseases are among the world's most prevalent public health problems, causing nearly 12 million deaths annually; about 25% of the deaths occur among individuals aged 25-69 years of age, as estimated by the World Health Organization (WHO). Urban areas have lost

more people (32.8%) to rural areas (6.9%), while the actual death toll globally from one or the other cardiovascular cause exceeds 80%. More than 23.6 million deaths have been estimated due to heart diseases for the year 2030. The process of diagnosis is very complicated due to high costs, delays, and unavailability of expert medical professionals; hence, these will not be available for most people, especially in low-income regions. Conventional diagnosis, such as ECG, Echocardiography, Blood Tests, has been generally helpful in identifying previous co-morbidities but does not predict the future occurrence of risk. Increasingly machine learning (ML) models like Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), LR, DT XGBoost are being used for heart disease prediction. According to a Cleveland and Stat log dataset study, an economical application of the SVM shows the highest accuracy, although more needs to be done in feature selection and data processing to improve its efficiency and reliability.

IV. PROPOSED SYSTEM

Amongst the increasing incidences of cardiovascular diseases (CVDs), which are one of the causes of sudden deaths worldwide, is the proposed system that looks into their prevention. Despite CVD prediction counting as one among the problems that clinicians deal with when analyzing clinical data, Machine Learning Algorithms (MLAs) have progressed to use historical clinical information for model development and prediction. This system proposes the use of the Random Forest and Support Vector Machine (SVM) in an early heart disease detection pipeline starting from data collection, preprocessing, feature extraction, train-test splitting, model training, evaluation, and, finally, prediction visualization. By leveraging the healthcare datasets, the model is able to find patient demographics, vital signs, cholesterol levels, and lifestyle choices to generate accurate predictions. This system architecture shown in figure [1] ensures the CVD detection in an efficient and scalable manner towards improvement of early diagnosis and patient care. Simultaneously, it adopts web-based access through Python Flask, making it simple for doctors to enter patient particulars and, within the least possible time, receive a real-time risk assessment and percentage-wise predictions.

A. SUPPORT VECTOR MACHINE

SVMs or Support Vector Machines are supervised learning models that perform classification and regression. SVMs find the hyperplane best separating the different classes. The closest data points to this line are called support vectors, and these determine the margin between the classes.

Classification Inside SVM Type 1
 In this type of SVM, training is intended to reduce the error

function very much to guarantee the best possible classifying performance.

$$\frac{1}{2} \mathbf{w}^T \mathbf{w} + c \sum_{i=1}^N \epsilon_i$$

Subject to the following constraints:

$$y_i(\mathbf{w}^T \phi(x_i) + b) \geq 1 - \xi_i \text{ and } \xi_i \geq 0, i = 1, \dots, N$$

Where C signifies the constant capacity, w the coefficient vector, b an additive constant, besides certain parameters employed for managing non-separable input data.

B. RANDOM FOREST

Random Forest is one of the very much commonly implemented supervised learning algorithms on deep learning. It is applicable for classification as well as regression. In short, ensemble methods combine several classifiers to approach more complex problems, thus rendering the best solution for the problem.

Random Forest Prediction:

Classification: The final output of the Random Forest model is determined by majority voting (or mode) of all the decision trees

$$y^{\wedge} = \text{mode}(f_1(x), f_2(x), \dots, f_K(x))$$

$$f_1(x), f_2(x), \dots, f_K(x)$$

are the predictions from each tree. Random Forest's Formula in General Terms

C. ARCHITECTURE DIAGRAM

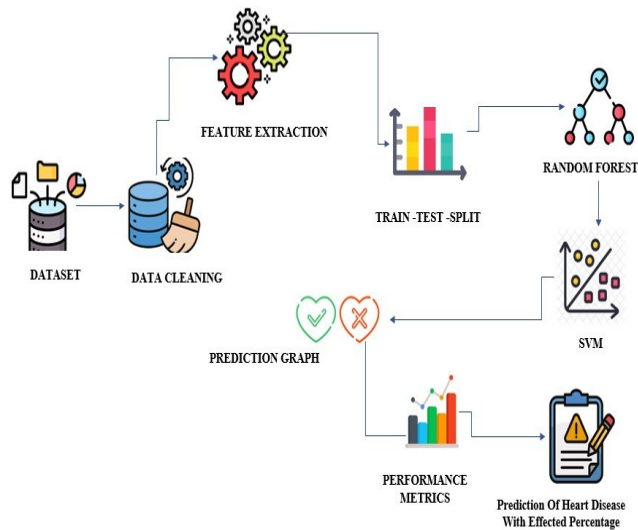


Fig. 1. Architecture diagram for proposed system

V. EXPERIMENTAL RESULTS

Heart disease prediction using the given dataset is done by employing algorithms like Random Forest and SVM. Implementation of the proposed model stems from the Python Flask web framework, evaluation of the model follows. Among the factors that increase heart disease risk are family history, smoking, unhealthy eating habits, high blood pressures, high cholesterol levels, overweight or obese, physically inactive lifestyle, and hypertension. Such parameters are considered crucial in the assessment of heart disease risk. Often, diagnosis arises from an amalgamation of current test results and the doctor's expertise. Given that the heart disease diagnosis is rather complicated, it requires profound experience and advanced medical knowledge.

A. DATASET

Dataset:

<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

This paper provides an open-access study on heart disease prediction, with a dataset of nearly 1,000 records. The dataset, which has been modified from the Statlog Heart Disease dataset, has been structured in three categories pertaining to heart disease. It comprises five key feature categories: key, input, and predictable attributes-play an important role within the predictive modeling process. Combined, they outline the study framework while also stressing its applicability in the area of heart disease classification.

Description of Dataset:

-Age of the patient: Age in years (as an integer).

-Sex: Gender of the patient (integer); 1 for male and 0 for female.

-Chest Pain Type (cp): Categorized into four integer types,
1: Typical angina
2: Atypical angina
3: Nonanginal pain
4: Asymptomatic

-Resting Blood Pressure: Measured in mm Hg (integer).

-Serum Cholesterol (chol): Cholesterol level in mg/dl (integer).

-Fasting Blood Sugar: It indicates the level of fasting blood sugar above 120 mg/dl.

(i) 1 if yes or 0 otherwise.

-Resting Electrocardiogram Results (restecg): (integer)

0: Normal

1: ST-T wave abnormality

2: Possible or definite left ventricular hypertrophy

Maximum Heart Rate Achieved (thalach): Recorded in beats per minute (integer).

-Exercise-Induced Angina (exang): (integer); 1 if present, 0 if absent.

-ST Depression (old peak): ST segment depression induced exercise-relation with rest (float).

-Slope of Peak Exercise ST Segment (slope): (integer)

1: Upsloping

2: Flat

3: Down sloping

-Number of Major Vessels (ca): Count of major vessels colored by fluorescein (float).

-Thalassemia Type (thal): (float)

3: Normal

6: Fixed defect

7: Reversible defect

-Heart Disease Diagnosis (num): Presence and severity of heart disease (integer):

0: No disease

1-4: Different severity levels of heart disease.

This model for analyzing prediction in heart diseases depends on the database.

B. IMPLEMENTATION MODULES

1. Data Collection

Heart disease datasets are information sets related to heart diseases, mainly used for analysis, machine-learning modeling, and other cardiovascular health research. Usually, these datasets come with a lot of features or attributes about people, like age, gender, blood pressure, cholesterol amounts, etc., together with diagnoses of heart disease.

2. Data Preprocessing

Preprocessing is known to prepare a dataset for a machine learning application by ensuring that the data is clean and structured correctly for the model learning algorithm. In heart disease prediction, preprocessing of the data involves handling missing values by encoding categorical variables; normalizing or scaling up to numerically valued features and settings up training and test sets. Different preprocessing methods such as normalization and missing value handling are used for refining data as shown in Figure [2].

3. Data Splitting

Data splitting is one of the important areas in machine learning that deals with the partitioning of the base data into training, validation, and test sets. Once again, this step helps to clearly evaluate the performance of the model and to ensure that the same would generalize adequately on new data as shown in Figure [3] and Figure [4].

4. Model Selection & Training

Setting a model and training it is an important part of the process to build a predictive system for the diagnosis of heart disease conditions using machine learning. After preprocessing the data (cleaning, transformation, and splitting into training and test sets), the next step is to select which machine learning model best predicts the likelihood of heart disease from a set of given features as shown in (Fig. 5, Fig. 6, Fig. 7, Fig. 8).

5. User Interaction with Models

User interfaces pertaining to this system are designed for ease of use so that healthcare providers can work seamlessly with the model. Once a patient has been entered into the system, a predictive result showing the chances of the patient suffering from a cardiovascular disease (CVD), together with a percentage of likelihood and the particular type of disease, will be generated. A flow diagram representing these processes is shown in Figure [9] and Figure [10].

C. COMPUTATIONAL COST ANALYSIS

The heart disease prediction model merged Random Forest and SVM classifiers into a combined heart disease predictive model. Random Forest is one of the techniques that are best for high dimensions dataset and performs better under parallel computation where the training complexity is $O(n \cdot m \cdot \log m)$. SVM is for a small, well-defined dataset, but it is very computationally intensive as the volume of data increases; its complexity is between $O(n^2)$ and $O(n^3)$. Although Random Forest is practically more efficient in terms of accuracy versus

computational time trade-offs, it can be feasibly executed using both algorithms on a typical desktop with (8 GB of RAM and an Intel i5 processor)

D. RESULT ANALYSIS MAIN PAGE

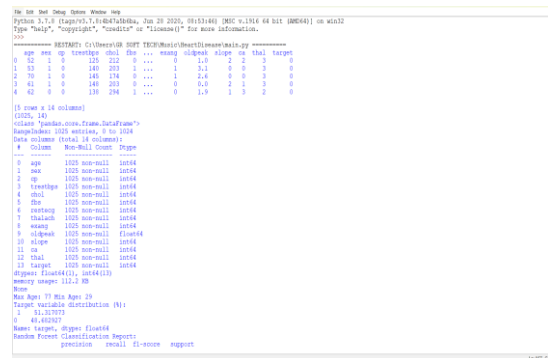


Fig 2: Dataset Pre-processing

Random Forest Classification Report:				
	precision	recall	f1-score	support
0	0.85	0.89	0.87	150
1	0.89	0.85	0.87	150
accuracy	0.87	0.87	0.87	300
macro avg	0.87	0.87	0.87	300
weighted avg	0.87	0.87	0.87	300

Support Vector Classification Report:				
	precision	recall	f1-score	support
0	0.87	0.74	0.80	150
1	0.78	0.99	0.83	150
accuracy	0.83	0.82	0.82	300
macro avg	0.83	0.82	0.82	300
weighted avg	0.82	0.82	0.82	300

SVM Classification Report:				
	precision	recall	f1-score	support
0	0.90	0.91	0.90	150
1	0.91	0.91	0.91	150
accuracy	0.91	0.91	0.91	300
macro avg	0.91	0.91	0.91	300
weighted avg	0.91	0.91	0.91	300

Model: Random Forest
Accuracy: 0.90
Precision: 1.00
Recall: 0.90

Fig 3: Algorithm Accuracy Report

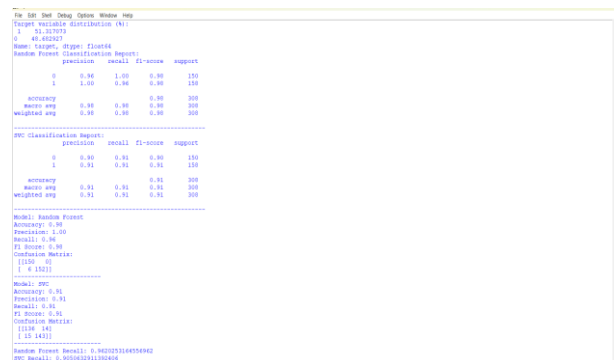


Fig 4: Algorithm Accuracy Report

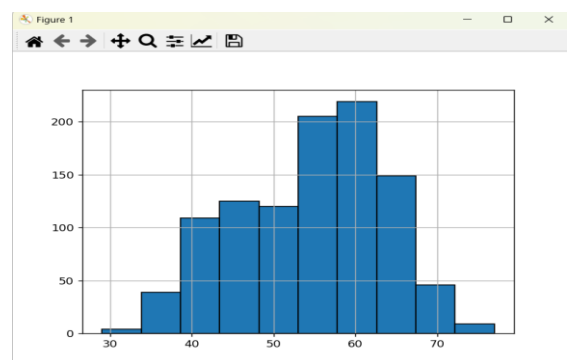


Fig 5: Heart Disease Report

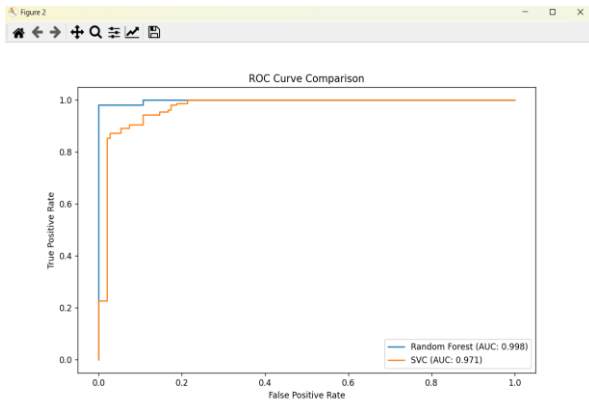


Fig 6: ROC Curve Comparison

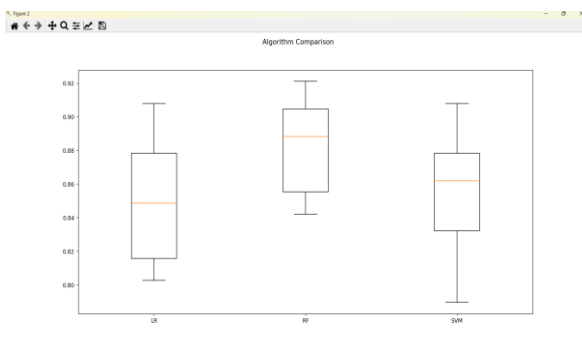


Fig 7: Accuracy Algorithm Chart

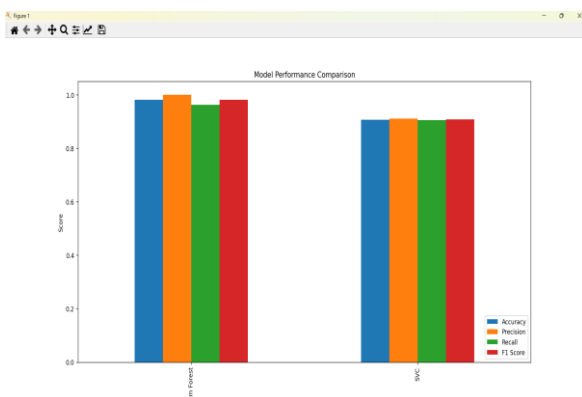


Fig 8: Accuracy Algorithm Chart

D. EVALUATION MATRIX

Accuracy: The number of correct predictions over the total number of predictions.

Precision and Recall: Positive predictions that are true regarding negative heart disease cases with respect to the ability of the model to predict.

F1 Score: is a number which weighs precision with respect to recall.

ROC curve and AUC: indicate how well the model distinguishes healthy patients from sick ones. Out of all techniques, Random Forest (RF) was very effective among them with an accuracy of 98%-which is the highest. Validation on ROC curve as shown in Fig [6] further justifies the superiority of RF-classifier.

This puts RF ahead as the most reliable and applicable in the real world with the capacity to deal with multi-dimensional complex data as shown in Table [1].

Table 1: Model Performance Analysis

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	82	0.78	0.89	0.82
KNN	87	0.89	0.85	0.87
SVM	91	0.91	0.91	0.91
Random Forest	98	0.96	1.00	0.98

MAIN PAGE

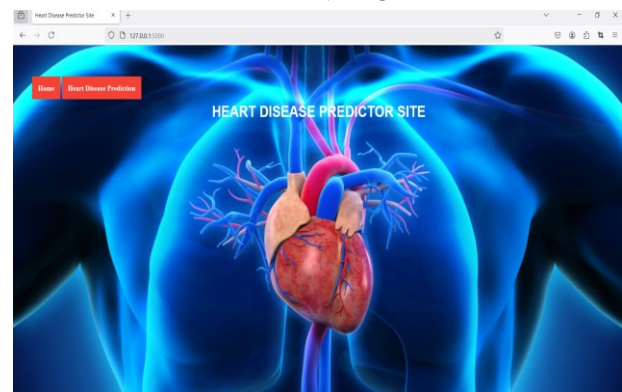


Fig 9: Main page

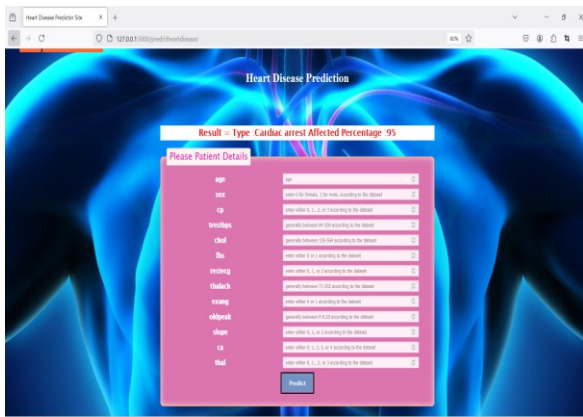


Fig 10: Disease Prediction

VI. CONCLUSION

Around 18 million people or about 7% of the Indian population suffer from heart diseases, of which more than half involve patients under the age of 65 years. Our research focused on diagnosing three major heart diseases: coronary artery disease, cardiovascular disease, and cardiomyopathy. In structuring the machine learning techniques around Random Forest and Support Vector Machine, the prime focus was to improve diagnosis accuracy. Random Forest can handle high-dimensional data, and evaluate vital features, whereas Support Vector Machine clearly demarcates strong decision boundaries for classification. Early diagnosis and treatment personalization provided in these frameworks with machine learning techniques does come with some limitations in practical applications. The study was limited in terms of clinical applicability with its model being validated on a very small dataset. Furthermore, the issues of real-time data and scalability in practical applications were neither discussed nor solved, which may create barriers to the deployment within the healthcare system. Future work will focus on deep learning models using larger and more diverse datasets while employing real-time data collection from wearable devices in the continuous monitoring of patients. Another appealing avenue is combining the RF, SVM, and XGBoost methods with neural networks to increase prediction accuracy. Additional health-related variables having genetic predisposition and lifestyle factors may also contribute greatly to improving the accuracy of treatment recommendations so that health professionals can tailor their services to patients.

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